

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

VII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
22PC1DS401	Data Science for BFSI	3	0	0	3	3
22PC1DS402	Deep Learning Applications	3	0	0	3	3
	PROFESSIONAL ELECTIVE – III					
22PE1DS401	Natural Language Processing and Text Analytics	3	0	0	3	3
22PE1CY301	Information and Network Security					
22PE1DS402	Distributed Computing					
22PE1CS401	Advanced Data Bases					
22PE1AM403	Information Retrieval Systems					
	PROFESSIONAL ELECTIVE – IV THROUGH SWAYAM-NPTEL					
	Machine Learning Stream	3	0	0	3	3
	Cyber Security Stream					
	Programming Stream					
	Artificial Intelligence Stream					
	Artificial Intelligence Stream					
	OPEN ELECTIVE – III	3	0	0	3	3
22PC2DS401	Data Science for BFSI Laboratory	0	0	2	2	1
22PC2DS402	Deep Learning Applications Laboratory	0	0	2	2	1
22PW4AD401	Major Project Phase – I	0	0	6	6	3
Total		15	0	10	25	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
ELA – Experiential Learning Assessment
LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

VIII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
	PROFESSIONAL ELECTIVE – V					
22PE1DS403	Advanced Data Science	3	0	0	3	3
22PE1CS403	Computer Vision					
22PE1AM405	Cognitive Engineering					
22PE1CS404	Augmented Reality and Virtual Reality					
22PC1CY401	Blockchain Technology					
	PROFESSIONAL ELECTIVE – VI					
22PE1AM302	Explainable AI	3	0	0	3	3
22PE1DS405	Time Series Analysis and Forecasting					
22PE1DS406	Quantum Computing					
22PE1AM407	Robotic Process Automation					
22PE1CS405	Social Media Analytics					
	OPEN ELECTIVE – IV	3	0	0	3	3
22PW4AD402	Major Project Phase – II	0	0	22	22	11
Total		9	0	22	31	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
ELA – Experiential Learning Assessment
LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1DS401) DATA SCIENCE FOR BFSI

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To explain the markets and several type of investors
- To understand about several terminologies related to investment
- To analyze the candle stick and channels for several types of markets
- To explain about several indicators related to markets
- To explain Wyckoff methodologies for investment

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the fundamentals of markets and investors

CO-2: Develop skills related to stock market terminologies

CO-3: Design the strategies for technical analysis

CO-4: Analyze indicators related to charts and patterns

CO-5: Summarize about Wyckoff methodologies

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	2	2	1	2	2	2	3	3	2	3
CO-2	3	3	3	2	3	2	2	1	1	2	2	3	3	2	3
CO-3	3	2	2	2	3	1	1	1	3	2	2	2	3	2	2
CO-4	3	2	3	3	3	2	2	1	2	2	2	3	3	3	3
CO-5	3	3	3	3	3	1	1	1	1	2	2	3	3	2	3

UNIT-I:

NSE, BSE, SEBI, RBI, NIFTY 50, NIFTY BANK, BSE SENSEX, Nifty VIX, Market capitalization, Large Cap, Mid Cap, Small Cap, Various kinds of Investors: promoters, FII, DII, NBF, Mutual Funds, public, create an effective portfolio, politics Vs. Stock market, RBI repo rates Vs Stock market, sectors and browse sectors in screener.in, John Bogle's five simple rules of investing, IPO.

UNIT-II:

Stock screener, Fundamental Analysis, Investment terminologies: Stock P/E, Industry P/E, P/E Ratio analysis, Price/Earning myth and reality, Revenue Growth, ROE, ROA, , ROCE, Debt to equity, Price to book value, price to book value myth and reality, Face Value, EBITDA, EPS, Earning Yield, Book Value, PEG Ratio, CAGR, Dividend Yield.

UNIT-III:

Technical Analysis, Trend lines, pivot points, candle stick chart, types of candle stick: Hanging man, Hammer, Bearish Engulfing pattern, Bullish Engulfing pattern, Doji. Rising window candle stick Patterns: Ascending channel pattern, Descending channel

pattern, Rectangular Channel pattern, Ascending Triangle pattern, Descending triangle pattern, symmetric triangle pattern, Flag pattern, pennant, head and shoulder pattern. Cup and handle pattern.

UNIT-IV:

Stock Indicators, commonly used moving averages, Alligator Indicator, Fear and Greed indicator, Moving averages, Ichimoku indicator, Fibonacci indicator, RSI indicator, stochastic indicator, MACD indicator, VWMA indicator, Summary of common stocks and uncommon profits, financial statements Review.

UNIT-V:

Spread Analysis, Single candle stick Wyckoff Trading Methodology, Breakdown of a single candle stick from Wyckoff Trading Methodology, Breakdown of Doji candle stick from Wyckoff trading methodology, How to analyze candle stick volume from Wyckoff trading methodology, price and volume in single candle stick from Wyckoff Trading, multiple candle stick analysis from Wyckoff, How to find shooting star candle stick traps and hammer candle stick traps from Wyckoff trading methodology.

TEXT BOOKS:

1. Machine Learning in Finance, M. Dixon, I Halperin and P. Bilokon, 1st Edition, Springer
2. Advances in Financial Machine Learning, Marcos Lopez, 1st Edition, Wiley
3. Machine Learning for Asset Managers, Marcos Lopez, 1st Edition, Cambridge University Press

REFERENCES:

1. Machine Learning for Algorithmic Trading, Stefan Jansen, 2nd Edition, Packt
2. Modern Portfolio Theory, Elton & Gruber, 9th Edition, Wiley
3. Investment Analysis and Portfolio Management, Reilly, Frank K., 5th Edition, Dryden

ONLINE RESOURCES:

1. <https://tradebrains.in/resources-stock-market-india/>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1DS402) DEEP LEARNING APPLICATIONS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To develop and apply major deep learning techniques, the problem settings, and their applications to solve real world problems
- To identify the deep learning algorithms which are more appropriate for various types of learning tasks in various domains
- To correlate and apply the comfortable of tools and techniques required in handling large amounts of datasets

COURSE OUTCOMES: After completion of the course, the student will be able to

CO1: Understand and apply the multi-layered perceptron for solving the stochastically complex problems

CO 2: Identify and evaluate the speed performance and dimension reduction using the SGD and auto encoder

CO 3: Apply and analyze various activation function with GPU in deep learning using Tensor flow and Keras

CO 4: Relate and build deep learning algorithms and solve real-world problems

CO 5: Analyze the case study of music generation system using deep learning

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	2	2	1	2	2	2	3	3	2	3
CO-2	3	3	3	2	3	2	2	1	1	2	2	3	3	2	3
CO-3	3	2	2	2	3	1	1	1	3	2	2	2	3	2	2
CO-4	3	2	3	3	3	2	2	1	2	2	2	3	3	3	3
CO-5	3	3	3	3	3	1	1	1	1	2	2	3	3	2	3

UNIT-I:

Diagrammatic representation Logistic Regression and Perceptron, Multi-Layered Perceptron (MLP) Notation, Training a single-neuron model, Training an MLP Chain Rule, training an MLP Memorization, back propagation, activation functions, vanishing gradient problem, bias variance trade-off, decision surfaces playground.

UNIT-II:

Deep Multi-layer perceptron's 1980s to 2010s, Dropout layers & Regularization, Rectified Linear Units (ReLU), Weight initialization, Batch Normalization, Optimizers Hill-descent analogy in 2D, Optimizers Hill descent in 3D and contours. SGD Recap, Batch SGD with momentum, Nesterov Accelerated Gradient (NAG)

UNIT-III:

Optimizers Ada Grad, Optimizers Ad delta and RMS Prop, Adam, Which algorithm to choose when, Gradient Checking and clipping, Softmax and Cross-entropy for multi-class classification, How to train a Deep MLP, Auto Encoders, Word2Vec CBOW, Word2Vec Skip-gram, Word2Vec Algorithmic Optimizations

UNIT-IV:

Tensor flow and Keras overview, GPU vs CPU for Deep Learning, Google Colaboratory, Install Tensor Flow, Online documentation, Softmax Classifier on MNIST dataset, MLP Initialization, Model 1 Sigmoid activation, Model 2ReLU activation, Model 3 Batch Normalization, Model 4 Dropout, MNIST classification in Keras, Hyper parameter tuning in Keras.

UNIT-V:

Case Study: Music Generation using Deep Learning:

Real-world problem-MUSIC, Music representation, Char-RNN with abc-notation Char-RNN model, Char-RNN with abc-notation Data preparation, Char-RNN with abc-notation Many to Many RNN,Time Distributed-Dense layer,Char-RNN with abc-notation State full RNN,Char-RNN with abc-notation Model architecture. Model training, Char-RNN with abc-notation music generation, Char-RNN with abc-notation with Tabla music, MIDI music generation, Survey blog.

TEXT BOOKS:

1. Deep Learning, Goodfellow, I., Bengio Y., Courville A., MIT Press, 2016
2. Deep Learning: A Practitioner's Approach, Josh Patterson, Adam Gibson, O'Reilly Media, 2017

REFERENCES:

1. Machine Learning with Tensor Flow, Nishant Shukla
2. Deep Learning with Keras, Antonio Gulli and Sujit Pal

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1DS401) NATURAL LANGUAGE PROCESSING AND TEXT ANALYTICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To create structured data out of free text content
- To implement recurrent network for language models
- To implement a PCA and ML algorithms in real life problems

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Extract information from text automatically using concepts and methods from natural language processing (NLP) including stemming, n-grams, POS tagging, and parsing

CO-2: Analyze the text syntactically and semantically

CO-3: Develop speech-based applications that use speech analysis (phonetics, speech recognition, and synthesis)

CO-4: Analyze large volume text data generated from a range of real-world applications

CO- 5: Analyze EDA, feature extraction, machine learning models, visualization, and text processing

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	2	2	1	2	2	2	3	3	2	3
CO-2	3	3	3	2	3	2	2	1	1	2	2	3	3	2	3
CO-3	3	2	2	2	3	1	1	1	3	2	2	2	3	2	2
CO-4	3	2	3	3	3	2	2	1	2	2	2	3	3	3	3
CO-5	3	3	3	3	3	1	1	1	1	2	2	3	3	2	3

UNIT-I:

Data set overview Amazon Fine Food reviews (EDA), Data Cleaning Deduplication, Why convert text to vector, Bag of words(BoW),Text Preprocessing Stemming, Stop-word removal, Tokenization, Lemmatization, uni-gram, bi-gram, n-grams.

UNIT-II:

TF-IDF, Why use log in IDF, Word2Vec, Avg-Word2Vec, TF-IDF weighted Word2Vec, Bag of Words (Code Sample), text processing (Code Sample), Bi-Grams and n-grams (Code Sample).

UNIT-III:

TF-IDF (Code Sample), Avg-Word2Vec and TF-IDF-Word2Vec (Code Sample), Word2Vec, Word2Vec (Code Sample), Avg-Word2Vec and TFIDF-Word2Vec (Code Sample).

UNIT-IV:

Case Study: Quora question pair similarity problem, How to optimally learn from case-studies, Business Real world problem definition, Business objectives and constraints, Mapping to a data science problem and data overview, Mapping to a data science problem and performance metric.

UNIT-V:

Train-test split, EDA Basic Statistics, EDA Basic Feature Extraction, EDA Text processing, EDA Advanced Feature Extraction Feature analysis Data Visualization by T-SNE, EDA TF-IDF weighted Word2Vec futurization.

Data science Models Loading Data, data science Models Random Model, data science Models Logistic Regression and Linear SVM, data science Models XGBoost.

TEXT BOOKS:

1. Deep Learning in Natural Language Processing, Li Deng, Yang Liu, Springer
2. Introduction to Natural Language Processing, Jacob Eisenstein, MIT Press

REFERENCES:

1. Transformers for Natural Language Processing, Denis Rothman, Packt Publishing
2. Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, 1st Edition, O'Reilly Media
3. Deep Learning for NLP and Speech Recognition, Uday Kamath, John Liu, James Whitaker, 1st Edition, Springer
4. Handbook of Natural Language Processing, Nitin Indurkha, Fred J. Damerau, 2nd Edition, Chapman and Hall/CRC
5. Speech and Language Processing, Daniel Jurafsky, James Martin, 2nd Edition, Press Prentice Hall

B.Tech. VII Semester

(22PE1CY301) INFORMATION AND NETWORK SECURITY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand various cryptographic algorithms
- To examine various public-key cryptosystems and the authentication techniques
- To apply the cryptographic primitives to build the secure protocols
- To analyze and design network security protocols
- To Identify and mitigate software security vulnerabilities in existing systems

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand basic cryptographic algorithms, message and web authentication and security issues

CO-2: Analyze the various authentication requirements and build secure protocols

CO-3: Analyze various cryptographic primitives and key distribution schemes

CO-4: Examine web security and wireless security

CO-5: Interpret E-mail security and analyze the security of transactions

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	-	-	1	-	1	-	1	-	-	3	2	2
CO-2	3	3	3	2	1	2	-	1	1	3	-	3	3	2	2
CO-3	3	3	1	2	1	1	-	1	1	2	1	2	3	3	2
CO-4	1	3	3	3	3	3	-	3	2	2	2	3	3	3	2
CO-5	1	3	3	2	3	3	-	3	2	2	2	3	3	3	2

UNIT-I:

Introduction: Overview and History of Cryptography, Public and Private-key Cryptography, One-Way Functions, Simple Encryption Schemes: Substitution and Transposition Techniques

Symmetric Key Ciphers: Block Cipher- Definition, Block Cipher Operations, DES, Triple DES, AES. Stream Ciphers- Definition, RC4, RC5.

UNIT-II:

Asymmetric Key Ciphers: Practical Aspects of Public-Key Cryptography, RSA Cryptosystem, Elgamal Cryptosystem, Elliptic Curve Cryptosystems, Merkle-Hellman Knapsack.

Data Integrity: Introduction, Hash Functions-Definition, SHA-512, Message Authentication Code: Authentication requirements, HMAC, CMAC, Digital signatures, Elgamal Digital Signature Scheme.

UNIT-III:

Key Management and Distribution: Symmetric Key Distribution Using Symmetric & Asymmetric Encryption, Distribution of Public Keys, Kerberos, X.509 Authentication Service, Public – Key Infrastructure

Advanced Topics: Pseudorandom Generators, Zero-knowledge Proofs.

UNIT-IV:

Transport-level Security: Web security considerations, Secure Socket Layer and Transport Layer Security, HTTPS, Secure Shell (SSH)

Wireless Network Security: Wireless Security, Mobile Device Security, IEEE 802.11 Wireless LAN, IEEE 802.11i Wireless LAN Security

UNIT-V:

E-Mail Security: Pretty Good Privacy, S/MIME IP Security: IP Security overview, IP Security architecture, Authentication Header, Encapsulating security payload, Combining security associations, Internet Key Exchange.

Case Studies on Cryptography and Security: Secure Multiparty Calculation, Virtual Elections, Single Sign On, Secure Inter-branch Payment Transactions.

TEXT BOOKS:

1. Cryptography and Network Security - Principles and Practice, William Stallings, 7th Edition, Pearson Education, 2017
2. Cryptography: Theory and Practice, Douglas Stinson, 3rd Edition, Chapman & Hall/CRC, 2011
3. Cryptography and Network Security, Atul Kahate, 3rd Edition, McGraw-Hill, 2005

REFERENCES:

1. Cryptography and Network Security, C. K. Shyamala, N. Harini, Dr. T. R. Padmanabhan, 1st Edition, Wiley, 2011
2. Cryptography and Network Security, Forouzan Mukhopadhyay, 3rd Edition, McGraw-Hill, 2015
3. Foundations of Cryptography, Oded Goldreich, Cambridge University Press, 2001
4. Information Security, Principles, and Practice, Mark Stamp, Wiley, 2011
5. Principles of Computer Security, W. M. Arthur Conklin, Greg White, Tata McGraw-Hill, 2016

B.Tech. VII Semester

(22PE1DS402) DISTRIBUTED COMPUTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce concepts related to distributed computing systems
- To focus on performance and flexibility issues related to systems design decisions
- To expose students to current literature in distributed systems
- To prepare students for a real-world distributed application design

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand models of distributed computing

CO-2: Analyze algorithms for coordination, communication and synchronization in distributed systems

CO-3: Analyze distributed shared memory models

CO-4: Design and implement distributed file systems

CO-5: Design distributed algorithms for handling deadlocks

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	1	1	-	-	-	-	-	-	-	1	3	1	2
CO-2	2	2	1	1	1	-	-	-	-	-	-	-	3	2	3
CO-3	3	3	2	2	2	-	-	-	-	-	-	-	3	2	2
CO-4	1	1	3	3	2	-	-	-	-	-	1	1	3	2	2
CO-5	-	-	1	1	2	-	-	-	-	-	-	2	3	3	2

UNIT-I:

Distributed Computing Introduction: Definition, Relation to parallel systems, synchronous vs asynchronous execution, design issues and challenges .A Model of Distributed Computations: A Model of distributed executions, Models of communication networks, Global state of distributed system, Models of process communication. Logical Time: Logical clocks, scalar time, vector time, Matrix time, virtual time, Physical clock synchronization – NTP.

UNIT-II:

Global State and Snapshot Recording Algorithms: System model, Snapshot algorithms for FIFO channels, Variations of Chandy-Lamport algorithm, Snapshot algorithms for non-FIFO channels, Snapshots in a causal delivery system, Monitoring global state, Necessary and sufficient conditions for consistent global snapshots, Finding consistent global snapshots in a distributed computation.

Message Ordering and Group Communication: Message ordering paradigms, Group communication, Causal order (CO), Total order, Multicast, Propagation trees for

multicast, Application-level multicast algorithms, Fault-tolerant group communication, Multicast algorithms at the network layer.

UNIT-III:

Termination Detection: System model of a distributed computation, Termination detection using distributed snapshots, weight throwing and spanning-tree-based algorithms, Message-optimal termination detection, and Termination detection in a general distributed computing model, Termination detection in the atomic computation model, Termination detection in a faulty distributed system.

Distributed Mutual Exclusion Algorithms: Lamport's algorithm, Ricart–Agrawala algorithm, Singhal's dynamic information-structure algorithm, Lodha and Kshemkalyani's fair mutual exclusion algorithm, Quorum-based mutual exclusion algorithms, Maekawa's algorithm, Agarwal–El Abbadi quorum-based algorithm, Token-based algorithms, Suzuki–Kasami's broadcast algorithm, Raymond's tree-based algorithm.

UNIT-IV:

Deadlock Detection in Distributed Systems: System model, Models of deadlocks, Knapp's classification of distributed deadlock detection algorithms, Mitchell and Merritt's algorithm for the single resource model, Chandy–Misra–Haas algorithm for the AND model, Chandy–Misra–Haas algorithm for the OR model, Kshemkalyani–Singhal algorithm for the P-out-of-Q model .

Distributed Shared Memory: Abstraction and advantages, Memory consistency models, Shared memory mutual exclusion, Wait-freedom, Register hierarchy and wait-free simulations, Wait-free atomic snapshots of shared objects

UNIT-V:

Check Pointing and Rollback Recovery: Introduction, Background and definitions, Issues in failure recovery, Checkpoint-based recovery, Log-based rollback recovery, Koo–Toueg coordinated check pointing algorithm, Juang–Venkatesan algorithm for asynchronous check pointing and recovery, Manivannan–Singhal quasi-synchronous check pointing algorithm, Peterson–Kearns algorithm based on vector time, Helary–Mostefaoui–Netzer–Raynal communication-induced protocol.

TEXT BOOKS:

1. Distributed Computing, Ajay D. Kshemakalyani, Mukesh Singhal, Cambridge University Press, 2008
2. Distributed Systems - Principles and Paradigms, Andrew S. Tanenbaum, Maarten Van Steen, Prentice Hall of India, 2004

REFERENCES:

1. Distributed Computing: Fundamentals, Simulations, and Advanced Topics, Hagit Attiya, Jennifer Welch, Wiley
2. Distributed Systems: Principles and Paradigms, Andrew S. Tanenbaum, Maarten van Steen, Prentice Hall of India

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1CS401) ADVANCED DATABASES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introducing distributed database management system and its design issues
- To exploring several algorithms for processing queries and be able to use them
- To describe the methods to translate complex conceptual data models into logical and physical database designs
- To demonstrating query optimization and its algorithms
- To enumerating the concepts behind distributed transaction processing

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand data processing and architecture of distributed database

CO-2: Apply partitioning and fragmentation techniques to the data distributed across the network

CO-3: Design and optimize query processing strategies on distributed fragments

CO-4: Explain transaction processing and concurrency control in distributed databases

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)			
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3	PSO-4
CO-1	3	1	1	-	-	1	2	-	-	-	1	1	2	2	2	1
CO-2	2	2	3	3	1	1	-	-	1	-	1	1	3	2	2	-
CO-3	3	3	3	3	3	2	2	-	2	-	2	3	3	3	2	2
CO-4	2	2	2	2	1	3	2	-	2	-	-	2	3	3	2	3

UNIT-I:

Introduction: Distributed Data Processing, Distributed Database System, Promises of DDBSs, Problem areas.

UNIT-II:

Distributed DBMS Architecture: Architectural Models for Distributed DBMS, DDMBS Architecture.

Distributed Database Design: Alternative Design Strategies, Distribution Design issues, Fragmentation, Allocation.

UNIT-III:

Query Processing and Decomposition: Query Processing Objectives, Characterization of query processors, layers of query processing, query decomposition, Localization of distributed data.

UNIT-IV:

Distributed Query Optimization: Query optimization, centralized query optimization, Distributed query optimization algorithms.

UNIT-V:

Transaction Management: Definition, properties of transaction, types of transactions. Distributed concurrency control: Serializability, concurrency control Mechanisms & Algorithms, Time stamped & Optimistic concurrency control Algorithms, Deadlock Management.

TEXT BOOKS:

1. Principles of Distributed Database Systems, M. Tamer OZSU and Patuck Valduriez, Pearson, 2001
2. Distributed Databases, Stefano Ceri and Willipse Pelagatti, McGraw-Hill

REFERENCES:

1. Database System Concepts, Henry F. Korth, A Silberchatz and Sudershan, McGraw-Hill
2. Database Management Systems, Raghuramakrishnan and Johhanes Gehrke, McGraw-Hill
3. Readings in Database Systems, M. Stonebraker, 2nd Edition, Morgan Kauffman, 1993

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1AM403) INFORMATION RETRIEVAL SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To describe the differences between repositories like Database Management Systems, Information retrieval systems and data warehouse
- To discover various pre-processing techniques which can apply on text documents to outline the structure of queries and documents
- To articulate fundamental functions used in information retrieval such as automatic indexing, abstracting, clustering and visualizing data
- To learn the important concepts, algorithms, and data/file structures that are necessary to specify, design, and implement Information Retrieval (IR) systems
- To explore various text search algorithms

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify and understand the relationships between various Repository Systems

CO-2: Apply knowledge of data structures and indexing methods in information retrieval Systems

CO-3: Implement various clustering, searching techniques and information visualization techniques

CO-4: Analyze clustering techniques and algorithms using evaluation measures

CO-5: Implement various text search algorithms

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	2	3	3	2	1	1	2	2	2	2	2	1
CO-2	3	3	3	3	3	2	2	2	2	2	2	2	3	3	2
CO-3	3	3	3	3	3	3	2	2	2	3	3	2	3	3	3
CO-4	3	3	3	3	3	3	2	2	2	3	3	2	3	3	3
CO-5	2	2	2	2	3	3	2	1	1	2	2	2	3	3	3

UNIT-I:

Introduction: Definition, Objectives, Functional Overview, Relationship to DBMS, Digital libraries and Data Warehouses, Information Retrieval System Capabilities – Search, Browse, Miscellaneous.

UNIT-II:

Cataloging and Indexing: Objectives, Indexing Process, Automatic Indexing, Information Extraction

Data Structures: Introduction, Stemming Algorithms, Inverted file structures, N – gram data structure, PAT data structure, Signature file structure, Hypertext data structure.

UNIT-III:

Automatic Indexing: Classes of automatic indexing, Statistical Indexing, Natural language, Concept indexing, Hypertext linkages.

Document and Term Clustering: Introduction, Thesaurus generation, Item clustering, Hierarchy of clusters.

UNIT-IV:

User Search Techniques: Search statements and binding, Similarity measures and ranking, Relevance feedback, Selective dissemination of information search, weighted searches of Boolean Systems, Searching the Internet and hypertext.

UNIT-V:

Text Search Algorithms: Introduction, Software text search algorithms, Hardware text search systems.

Information System Evaluation: Introduction, Measures used in system evaluation, Measurement example – TREC results.

TEXT BOOKS:

1. Information Storage and Retrieval Systems: Theory and Implementation, Kowalski, Gerald, Mark T. Maybury, Springer
2. Modern Information Retrieval: Ricardo Baeza, Yates, Pearson Education, 2007

REFERENCES:

1. Information Retrieval: Algorithms and Heuristics, David A. Grossman and Ophir Frieder, 2nd Edition, Springer
2. Information Retrieval Data Structures and Algorithms, Frakes W. B., Ricardo Baeza-Yates, Prentice Hall, 1992

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2DS401) DATA SCIENCE FOR BFSI LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To explain about markets and several types of investors
- To understand about several terminologies related investment
- To analyze about candle stick and channels for several types of markets
- To explain about several indicators related to markets
- To explain Wyckoff methodologies for investment

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze financial analysis using real time data and visualization tools

CO-2: Develop expertise in financial screening and technical analysis using advanced tools for strategic stock investments

CO-3: Empower with advanced financial analysis expertise for strategic investment decisions

CO-4: Develop the technical analysis and data science skills for strategic investments

CO-5: Applying technical analysis tools and indicators to make strategic investment choices

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	3	2	3	1	1	1	2	2	2	2	3	2	1
CO-2	3	3	3	2	3	2	2	1	2	2	2	2	2	3	2
CO-3	3	2	2	2	3	1	1	1	2	2	2	2	3	2	2
CO-4	3	2	3	2	3	2	2	1	2	2	2	3	3	2	2
CO-5	3	3	3	3	3	2	2	1	2	2	2	3	3	3	2

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

1. Visit bseindia.com, nseindia.com and analyze graphs of BSE SENSEX 50 and BSE BANKEX

Identify Large Cap, Mid Cap. Small Cap companies based on Market Cap in screener.in

Check FII inflow increasing or decreasing, Check What FIIs are buying

Write a query on Change in FII holding greater than 1% AND (FII holding - Change in FII holding) less than 5% in screener.in

Write a query on Market capitalization > 500 AND
Price to earning < 15 AND Return on capital employed > 22%

Analyze Major Global Indices in investing.com

Analyze stock heat map in trading view.com

Construct heat map for sample data from screener.in

2. Browse about screener.in, moneycontrol.com,sgxnifty.com, investorgain.com, marketwatch.com, tickertape.in, tradingview.com, scanners.streak.com, trendlyne.com and chartink.com

Write a query on Market capitalization > 500 AND Price/ Earning < 15 AND ROCE > 22%

Write a query on Market Capitalization / Average Earnings 10Year < 15 AND Average dividend payout 3years > 20 AND Debt to equity < .2 AND Average return on capital employed 7Years > 20 in screener.in

Analyze price chart for 1 month, 6 months, 1 year, 3 years, 5 years, 10 years with price on NSE, 50 DMA, 200 DMA and volume in screener in for large cap, mid cap and small cap.

Build price chart for sample data for sample data from screener.

3. Write a query on YOY Quarterly sales growth > 15 and YOY Quarterly profit growth > 20 and Net Profit latest quarter > 1

Write a query on $100 * ((\text{High price} - \text{Current price}) / \text{High price}) < 10$ AND $100 * ((\text{High price} - \text{Current price}) / \text{High price}) > 0$ AND $100 * (\text{Current price} / \text{Low price} - 1) > 100$ AND Current price > 10 AND Volume > 100000

Analyze sales and margin chart for 1 month, 6 months, 1 year, 3 years, 5 years, 10 years with price on NSE, 50 DMA, 200 DMA and volume in screener in for large cap, mid cap and small cap.

Build sales and margin chart for sample data for sample data from screener.

4. Browse sector and market cap in stock universe of tickertape.in
Check profitability analysis for small cap, mid cap and large cap with respect to close price and PE ratio.

Check price and volume analysis for small cap, mid cap and large cap with respect to close price and PE ratio.

Check financial ratios, growth, valuation and ownership analysis for small cap, mid cap and large cap with respect to close price and PE ratio.

Write a query on DMA 50 > DMA 200 AND DMA 50 previous day < DMA 200 previous day

Analyze PE ratio chart for 1 month, 6 months, 1 year, 3 years, 5 years, 10 years with price on NSE, 50 DMA, 200 DMA and volume in screener in for large cap, mid cap and small cap.

Build PE ratio chart for sample data for sample data from screener.

5. Check technical indicators, income statement, balance sheet and cash flow statement analysis for small cap, mid cap and large cap with respect to close price and PE ratio.

Write a query on DMA 50 > DMA 200 AND DMA 50 previous day < DMA 200 previous day

Analyze EV/EBITDA chart for 1 month, 6 months, 1 year, 3 years, 5 years, 10 years with price on NSE, 50 DMA, 200 DMA and volume in screener in for large cap, mid cap and small cap.

Build EV/EBITDA chart for sample data for sample data from screener.

6. Identify candle stick chart in chartink.com for small cap, mid cap and large cap.

Write a query on Stocks nearing 200 day high by 5% & showing consolidation for the past 30 days with volume above 50 day average.

Analyze about documents of large cap, mid cap and small cap in screener.in and interpret with appropriate Data science techniques.

7. Draw trend lines and identify pivot points of small cap, mid cap and large cap in chartink.com

Write a query on Stocks closing crossing the cloud & span A greater than span B, indicating a positive trend in chartink.com

Analyze Price to book chart for 1 month, 6 months, 1 year, 3 years, 5 years, 10 years with price on NSE, 50 DMA, 200 DMA and volume in screener in for large cap, mid cap and small cap.

Analyze about peers and quarter of a large cap, mid cap and small cap and apply appropriate data science technique to analyze this data.

8. Identify Hanging man, Hammer, Bearish Engulfing pattern, Bullish Engulfing pattern for different types of companies in chartink.com

Write a query on Bullish reversal pattern in which a stock which had a long white body a 2 days ago, then opened lower with a Doji a day ago and finally closed above the previous day.

Analyze Market cap/Sales chart for 1 month, 6 months, 1 year, 3 years, 5 years, 10 years with price on NSE, 50 DMA, 200 DMA and volume in screener in for large cap, mid cap and small cap.

9. Identify Doji. Rising window candle stick for different types of companies.

Analyze about profit, loss and balance sheet of a large cap, mid cap and small cap and apply appropriate data science technique to analyze this data.

10. Identify Ascending channel pattern, Descending channel pattern, Rectangular Channel pattern, Ascending Triangle pattern in chartink.com
11. Identify Descending triangle pattern, symmetric triangle pattern, Flag pattern, pennant, head and shoulder pattern. Cup and handle pattern in Chartink.com
12. Identify subsectors in tickertape.in with respect to market cap and close price commonly used moving averages, Alligator Indicator in chartink.com
13. Fear and Greed indicator, Moving averages, Ichimoku indicator, Fibonacci indicator, RSI indicator, stochastic indicator, MACD indicator, VWMA indicator Wyckoff Trading Methodology.

TEXT BOOKS:

1. Machine Learning in Finance, M. Dixon, I. Halperin and P. Bilokon, 1st Edition, Springer
2. Advances in Financial Machine Learning, Marcos Lopez, 1st Edition, Wiley
3. Machine Learning for Asset Managers, Marcos Lopez, 1st Edition, Cambridge University Press

REFERENCES:

1. Machine Learning for Algorithmic Trading, Stefan Jansen, 2nd Edition, Packt
2. Modern Portfolio Theory, Elton & Gruber, 9th Edition, Wiley
3. Investment Analysis and Portfolio Management, Frank K., 5th Edition, Dryden

ONLINE RESOURCES:

1. <https://tradebrains.in/resources-stock-market-india/>

B.Tech. VII Semester

(22PC2DS402) DEEP LEARNING APPLICATIONS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To understand the fundamental principles of autonomous vehicles
- To develop skills to both design and critique visualizations
- To obtain a comprehensive knowledge of various techniques for deep learning
- To analyse the various case studies of data analysis

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the fundamentals of autonomous vehicles like self-driving cars

CO-2: Learn the basics of deep learning techniques and how they are applied to HAR

CO-3: Equip students with the knowledge and skills necessary to understand and solve the Quora Question Pair Similarity problem

CO-4: Grasp the foundational concepts of music generation using deep learning techniques

CO-5: Learn to design the models of the problem statements using RNN & CNN

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	2	1	1	2	2	2	2	3	2	2
CO-2	3	3	3	3	2	2	2	1	1	2	2	3	3	2	3
CO-3	2	2	2	2	3	1	1	1	2	2	2	2	3	2	2
CO-4	2	3	2	2	3	1	2	1	2	2	2	3	3	3	3
CO-5	3	2	3	3	3	1	1	1	1	2	2	3	3	2	3

LIST OF PROGRAMS / EXERCISES:

WEEK 1-4: Self Driving Car in Deep Learning

- Problem definition and dataset
- Data understandings analysis files and folders
- Dash-com, images and steering analysis
- EDA steering angles
- Mean baseline model
- Deep learning model, Deep learning for regression
- Batch load the dataset
- NVIDIA's end to end CNN MODEL
- Train the model
- Test and visualize the output and its extensions

WEEK 5-8: Human Activity Recognition

- a) Problem definition
- b) Dataset understanding
- c) Data cleaning and pre-processing
- d) EDA univariate analysis
- e) Classical ML models
- f) Deep learning models

WEEK 9-12: Quora Question Pair Similarity Problem

- a) Business real world problem definition
- b) Business objectives and constraints
- c) Mapping to an ML problem data overview
- d) Mapping to an ML problem and performance measures
- e) Mapping to an ML problem train-test split
- f) EDA basic statistics, feature extraction
- g) EDA text processing, advanced feature extraction
- h) EDA feature analysis, data visualization
- i) EDA IT-IDF weighting word2vec factorization
- j) ML models loading, random model
- k) ML models logistics regressions and linear SVM

WEEK 13-14: Music Generation Using Deep Learning

- a) Real-world problem and music representation
- b) Char-RNN with abc –notation char CNN model
- c) Char-RNN with abc –notation data preparation
- d) Char-RNN with abc –notation many to many RNN time distribution
- e) Char-RNN with abc –notation state full RNN,model architecture
- f) Char-RNN with abc –notation music generation, generate tabla music.
- g) MIDI music generation.

WEEK 15: LAB INTERNAL.**TEXT BOOKS:**

1. Computational and Inferential Thinking: The Foundations of Data Science, Adi Adhikari and John De Nero, 1st Edition, 2019
2. Doing Data Science, Straight Talk from The Frontline, Cathy O'Neil and Rachel Schutt, O'Reilly, 2014
3. Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures, Claus Wilke, 1st Edition, O'Reilly, 2019

REFERENCES:

1. Mining of Massive Datasets, v2.1, Jure Leskovek, Anand Rajaraman and Jeffrey Ullman, Cambridge University Press, 2014
2. Practical Statistics for Data Scientists: 50 Essential Concepts, Bruce, Peter, and Andrew Bruce, O'Reilly, 2017
3. Data Analysis and Visualization Using Python: Analyze Data to Create Visualizations for BI Systems, Ossama Embarak, Press, 2018

B.Tech. VIII Semester

(22PE1DS403) ADVANCED DATA SCIENCE

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To create structured data out of free text content
- To implement recurrent network for language models
- To implement a PCA and ML algorithms in real life problems

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Identify and understand the local minima and maxima by using the Gradient descent algorithm

CO-2: Apply and analyze the filter effects of images using the concept of convolution and data augmentation

CO-3: Understand & validate the use of text processing and time series data with LSTM and RNN and optimizing the code with transfer learning

CO-4: Demonstrate and measure the real time scenario (Case study) by using deep learning & Analytics techniques

CO 5: Analyze the case study of self-driving car using CNN models

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	2	2	1	2	2	2	3	3	1	2
CO-2	3	3	3	2	3	2	2	1	1	2	2	3	3	1	3
CO-3	3	2	2	2	3	1	1	1	3	2	2	2	3	2	3
CO-4	3	2	3	3	3	2	2	1	2	2	2	3	3	2	3
CO-5	3	3	3	3	3	1	1	1	1	2	2	3	3	1	2

UNIT-I:

Maxima and Minima, Vector calculus Grad, Gradient descent geometric intuition, Learning rate, Gradient descent for linear regression, SGD algorithm, Constrained Optimization & PCA, Logistic regression formulation revisited.

UNIT-II:

Convolution Edge Detection on images, Convolution Padding and strides, Convolution over RGB images, Convolutional layer, Max-pooling, CNN Training Optimization.

UNIT-III:

Receptive Fields and Effective Receptive Fields, Example CNN LeNet [1998], ImageNet dataset, Data Augmentation, Convolution Layers in Keras, AlexNet, VGGNET, Residual Network.

UNIT-IV:

Inception Network, What is Transfer learning, Code example Cats vs Dogs, Code Example MNIST dataset. Why RNNs, Recurrent Neural Network, Training RNNs Backprop, Types of RNNs, Need for LSTMGRU, LSTM, GRUs, Deep RNN, Bidirectional RNN, Code example IMDB Sentiment classification,

UNIT-V:

Case study on Real World Cases: Self Driving Car, Self Driving Car Problem definition. SDC, Datasets, Data understanding & Analysis Files and folders, Dash-cam images and steering angles.

EDA Steering angles, Mean Baseline model simple, Deep-learning model, Deep Learning for regression CNN, CNN+RNN, Batch load the dataset, NVIDIA's end to end CNN model, Train the model, Test and visualize the output, Extensions.

TEXT BOOKS:

1. Deep Learning, Ian Goodfellow, Yoshua Bengio, Aaron Courville, MIT Press, 2017
2. Deep Learning: A Practitioner's Approach, Josh Patterson, Adam Gibson, O'Reilly, 2017

REFERENCES:

1. Neural Networks and Deep Learning, Michael Nielsen, Determination Press, 2015
2. Deep Learning with Keras, Antonio Gulli and Sujit Pal, 2017

B.Tech. VIII Semester

(22PE1CS403) COMPUTER VISION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce the basic concepts of computer vision
- To explore the low-level feature extraction methods
- To understand the deep learning models for computer vision
- To introduce 3D computer vision and its applications

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Describe the basic computer vision concepts

CO-2: Extract the feature extraction methods

CO-3: understand the deep learning models for computer vision

CO-4: Explore and identify 3D computer vision applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	1	1	1	2	1	1	1	1	1	1	1	1	1
CO-2	2	3	3	3	3	3	1	1	1	1	1	1	3	3	3
CO-3	3	3	3	3	3	3	1	1	1	1	1	1	3	3	3
CO-4	1	1	1	1	1	2	1	1	1	1	1	1	1	2	1

UNIT-I:

Digital Image Fundamentals: Elements of visual perception, light and the electromagnetic spectrum, Image sensing, and Acquisition, Sampling and Quantization, Basic relationships between pixels, Gray level transformations, Histogram processing, Spatial filtering: smoothing and sharpening.

UNIT-II:

Visual Feature Extraction: Low-level Feature Extraction: Color histograms, First order edge detection, Second order edge detection, Scale Invariant Feature Transform (SIFT), Texture features, Mathematical morphology.

UNIT-III:

Recognition: Image Recognition, Image classification, Object detection, semantic segmentation, Video understanding, vision, and language.

UNIT -IV:

Applications: Object Detection, Surveillance application, Face recognition, Activity recognition, Segmentation in autonomous driving, Robot vision

UNIT-V:

3D Computer vision: Stereo-pair images and depth perception, 3D vision systems, 3D vision applications, Case examples: 3D systems for vision impaired persons, face and body modeling, clinical and veterinary applications, movie restoration.

TEXTBOOKS:

1. Digital Image Processing, Gonzalez R. C., Pearson Education, 2009
2. Feature Extraction and Image Processing for Computer Vision, Nixon M. & Aguado A., Academic Press, 2019
3. Deep Learning in Computer Vision: Principles and Applications, Hassaballah M., & Awad A. I. (Eds.), CRC Press, 2020

REFERENCES:

1. An Introduction to 3D Computer Vision Techniques And Algorithms, Cyganek B., & Siebert J. P., John Wiley & Sons, 2011
2. Computer Vision: Algorithms and Applications, Szeliski R., Springer Nature, 2022

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1AM405) COGNITIVE ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To identify the basics of cognitive science engineering with focus on mind organization, brains
- To analyze human intelligence and artificial intelligence
- To understand basics of neuroscience and linguistics
- To analyze various levels of cognitive functioning and its applications
- To relate the concepts of cognitive field in robotics and real time applications

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Introduce role of computers in cognitive science and neuroscience applications

CO-2: Illustrate various information processing models and cognitive architecture

CO-3: Apply various computation tools for cognitive functioning

CO-4: Analyze cognitive science applications in image processing linguistic applications

CO-5: Demonstrate the use of cognitive models and patterns in distributed environment, and real time applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	1	2	1	1	1	2	1	1	1	2	2	2
CO-2	2	3	2	2	3	1	1	2	2	2	1	1	3	3	2
CO-3	2	3	3	2	2	1	2	1	2	2	2	1	3	3	3
CO-4	3	3	3	2	2	2	1	2	3	3	2	2	3	3	2
CO-5	2	2	2	3	2	1	2	1	2	2	3	2	2	2	3

UNIT-I:

Introduction to Cognitive Science: The Cognitive view –Some Fundamental Concepts – Computers in Cognitive Science – Applied Cognitive Science – The Interdisciplinary Nature of Cognitive Science, The Architecture of the Mind. Cognitive Processes, Computational Neuroscience - The Organization of the mind.

UNIT-II:

Higher-Level Cognition and Challenges: Dynamical systems and situated cognition- Challenges – Emotions and Consciousness – Computation of Cognitive Functioning in machines: Robotics, Human-Robotics Interaction.

UNIT-III:

Cognitive Science for Vision/Image Processing: Perception and sensing: visual cognition, cognitive mechanisms of vision, Feature Extraction from Images, Information Processing in Perception and Visual Behavior, Cognitive function measurement tools and software.

UNIT-IV:

Language Acquisition and Semantics: Milestones in acquisition, Theoretical perspectives, Semantics and Cognitive Science, Meaning and entailment, Sense, Problems in possible worlds Semantics, Cognitive and computational models of semantic Processing.

UNIT-V:

Distributed Cognitive Learning Using Fuzzy Cognitive Maps: Axelrod's cognitive maps, kosko's model, adaptive FCM, Pal and Konar's FCM model-examples.

TEXT BOOKS:

1. Cognitive Science: An Introduction, Neil Stillings, Steven E. Weisler, Christopher H. Chase and Mark H. Feinstein, 2nd Edition, 1995
2. Cognitive Science: An Introduction to the Science of the Mind José Luis Bermúdez, Cambridge University Press, 2010
3. Cognitive Engineering: A Distributed Approach to Machine Intelligence (Advanced Information and Knowledge Processing), Amit Konar, Springer, 2005

REFERENCES:

1. Computational Vision: Information Processing in Perception and Visual Behavior, Hanspeter A. Mallot, Translated by John S. Allen
2. How the Mind Works, Steven Pinker, 2009
3. Cognitive Science: An Interdisciplinary Approach, Carolyn Panzer Sobel and Paul Li, 2013
4. Mind: Introduction to Cognitive Science, Paul Thagard, 2nd Edition, MIT Press, 2004
Wilson Robert A., & Keil Frank C. (Eds.)
5. The MIT Encyclopedia of the Cognitive Sciences (MITECS), MIT Press, 2001

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CS404) AUGMENTED REALITY AND VIRTUAL REALITY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Programming and Data Structures

COURSE OBJECTIVES:

- To understand the basic concepts of Virtual and Augmented Reality
- To identify the differences in AR/VR concepts and technologies
- To understand the fundamental concepts relating to Virtual Reality such as presence, immersion, and engagement
- To evaluate usability of AR/VR applications and critique their use of AR/VR capabilities
- To design and prototype effective AR/VR applications using UNITY platform for business, industry, non-profit and government organizations

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Introduce role of augmented and virtual reality in interactive application development

CO-2: Apply Geometric Modeling Techniques

CO-3: Implement virtual environment with respect to user experience

CO-4: Design Virtual/Augmented Reality Applications

CO-5: Acquire fundamental and hands-on skills needed to develop AR/VR applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	1	1	1	1	2	2	1	-	2	2	3	3	3
CO-2	1	1	1	3	1	2	2	1	2	1	1	2	3	3	3
CO-3	1	1	2	-	3	1	1	1	1	2	1	2	3	3	3
CO-4	-	2	-	-	-	2	-	2	1	1	3	1	3	3	3
CO-5	-	-	2	-	3	-	-	-	-	-	-	-	3	3	3

UNIT-I:

Introduction to Augmented Reality (AR): Definition and Scope A Brief History of Augmented Reality Displays (Multimodal Displays, Spatial Display Model, and Visual Displays) Strong vs Weak AR Applications of AR Challenges in AR

UNIT-II:

Introduction to Virtual Reality (VR): Definition and Scope Types of VR Characteristics of VR Basic VR environments Limitations of VR environments Immersion Vs Presence Key hardware requirements for VR

UNIT-III:

Interaction Design for AR/VR Environments: Interaction design process Identifying user needs AR/VR design considerations Typical AR/VR Interface Metaphors Affordances in AR/VR Human Information Processing Design for Perception and cognition User experience (UX) guidelines for AR/VR UX challenges for AR/VR Prototyping for AR/VR Evaluation of the developed AR/VR prototype

UNIT-IV:

Unity Overview: Windows, Interface, Navigation, Terminology, Game Objects, Hierarchy, Parenting Objects Asset Store, Importing Plug-ins Creating a Terrain, Materials, Colors, Transparency Introduction to MonoBehaviour: Awake, Start, Update

UNIT-V:

Introduction to Vuforia and Physics in UNITY

Vuforia Overview: Interface, Navigation, Terminology, Image Targeting, Custom Images Overview of Physics in Unity

Introduction to Scripting: Terminology, Creating Objects, Accessing Components, Debugging, Lists, Loops

TEXT BOOKS:

1. Understanding virtual reality, Sherman W. R., & Craig A. B., Morgan Kaufman, 2003
2. Augmented Reality: Principles and Practice, Schmalstieg D., & Hollerer T., Addison-Wesley Professional, 2016

REFERENCES:

1. A survey of augmented reality. Presence: Teleoperators & Virtual Environments, Azuma R. T., 1997, 6(4), 355–385
2. Recent Advances in Augmented Reality, IEEE Computer Graphics and Applications, Azuma R., Baillot Y., Behringer R., Feiner S., Julier S., & MacIntyre B., 2001, 21(6), 34-47
3. A Cost-Effective Interactive 3D Virtual Reality System Applied To Military Live Firing Training, Virtual Reality, Bhagat K. K., Liou, W. K., & Chang, C. Y, 2016, 20(2), 127-140. doi:10.1007/s10055-016-0284-x
4. Augmented Reality Technologies, Systems And Applications, Multimedia Tools and Applications, Carmigniani J., Furht B., Anisetti M., Ceravolo P., Damiani E., & Ivkovic M., 2011, 51(1), 341-377
5. Human Augmentation: Past, Present and Future. International Journal Of Human Computer Studies, Raisamo R., Rakkolainen I., Majaranta P., Salminen K., Rantala J., & Farooq, 2019, 131, 131-143. doi: <https://doi.org/10.1016/j.ijhcs.2019.05.008>

ONLINE RESOURCES:

1. Virtual Reality Engineering, Manivannan M., 2018, IIT Madras, <https://nptel.ac.in/courses/121106013>
2. Industry 4.0: Augmented Reality and Virtual Reality, Misra S., 2019, IIT Kharagpur, <https://www.youtube.com/watch?v=zLMgdYI82IE>
3. Augmented Reality - Fundamentals and Development, NPTEL Special Lecture Series, Dube A., 2020, <https://www.youtube.com/watch?v=MGuSTAqIZ9Q>
4. <http://cambum.net/course-2.htm>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PC1CY401) BLOCKCHAIN TECHNOLOGY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To provide an understanding of blockchain technology, including its origin, components, types, consensus mechanisms, and cryptocurrency applications
- To explore public blockchain systems and smart contracts, their principles, implementations, and industry applications
- To know the security issues of blockchain technology
- To understand security challenges in blockchain and explore industry applications.
- To study practical blockchain implementations through case studies

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze blockchain technology by examining its origin, components, types, consensus mechanisms, and cryptocurrency applications

CO-2: Evaluate public blockchains and smart contracts, their principles, implementations, and industry applications

CO-3: Examine private and consortium blockchains, their applications, and the role of ICOs in blockchain fundraising

CO-4: Apply blockchain security principles and explore its applications, limitations, and future scope

CO-5: Implement blockchain solutions through case studies and development with Python and Hyperledger Fabric

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)			
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3	PSO-4
CO-1	2	2	2	1	1	1	1	2	1	1	1	2	2	2	2	1
CO-2	3	2	3	3	3	2	2	3	2	1	3	3	3	3	2	1
CO-3	2	2	2	1	1	1	1	2	1	1	1	2	3	3	3	-
CO-4	3	3	2	1	1	1	1	2	1	1	2	3	3	3	2	3
CO-5	3	3	3	1	1	1	1	1	1	1	1	2	2	2	3	2

UNIT-I:

Fundamentals of Blockchain: Introduction, Origin of Blockchain, Blockchain Solution, Components of Blockchain, Block in a Blockchain, The Technology and the Future. Blockchain Types and Consensus Mechanism: Introduction, Decentralization and Distribution, Types of Blockchain, Consensus Protocol. Cryptocurrency – Bitcoin, Altcoin and Token: Introduction, Bitcoin and the Cryptocurrency, Cryptocurrency Basics, Types of Cryptocurrencies, Cryptocurrency Usage.

UNIT-II:

Public Blockchain System: Introduction, Public Blockchain, Popular Public Blockchains, The Bitcoin Blockchain, Ethereum Blockchain. Smart Contracts: Introduction, Smart Contract, Characteristics of a Smart Contract, Types of Smart Contracts, Types of Oracles, Smart Contracts in Ethereum, Smart Contracts in Industry.

UNIT-III:

Private Blockchain System: Introduction, Key Characteristics of Private Blockchain, Why We Need Private Blockchain, Private Blockchain Examples, Private Blockchain and Open Source, E-commerce Site Example, Various Commands (Instructions) in E-commerce Blockchain, Smart Contract in Private Environment, State Machine, Different Algorithms of Permissioned Blockchain, Byzantine Fault, Multichain.

Consortium Blockchain: Introduction, Key Characteristics of Consortium Blockchain, Why We Need Consortium Blockchain, Hyperledger Platform, Overview of Ripple, Overview of Corda. Initial Coin Offering: Introduction, Blockchain Fundraising Methods, Launching an ICO, Investing in an ICO, Pros and Cons of Initial Coin Offering, Successful Initial Coin Offerings, Evolution of ICO, ICO Platforms.

UNIT-IV:

Security in Blockchain: Introduction, Security Aspects in Bitcoin, Security and Privacy Challenges of Blockchain in General, Performance and Scalability, Identity Management and Authentication, Regulatory Compliance and Assurance, Safeguarding Blockchain Smart Contract (DApp), Security Aspects in Hyperledger Fabric.

Applications of Blockchain: Introduction, Blockchain in Banking and Finance, Blockchain in Education, Blockchain in Energy, Blockchain in Healthcare, Blockchain in Real-estate, Blockchain in Supply Chain, The Blockchain and IoT. Limitations and Challenges of Blockchain.

UNIT-V:

Blockchain Case Studies: Case Study 1 – Retail, Case Study 2 – Banking and Financial Services, Case Study 3 – Healthcare, Case Study 4 – Energy and Utilities.

Blockchain Platform using Python: Introduction, Learn How to Use Python Online Editor, Basic Programming Using Python, Python Packages for Blockchain. Blockchain platform using Hyperledger Fabric: Introduction, Components of Hyperledger Fabric Network, Chain codes from Developer.ibm.com, Blockchain Application Using Fabric Java SDK.

TEXT BOOK:

1. Block Chain Technology, Chandramouli Subramanian, Asha A. George, Abhilash K. A. and Meena Karthikeyan, Universities Press

REFERENCES:

1. Blockchain Blueprint for Economy, Melanie Swan, SPD O'Reilly
2. Blockchain for Business, Jai Singh Arun, Jerry Cuomo, Nifin Gaur, Pearson Addison Wesley

ONLINE RESOURCES:

1. The Bitcoin Backbone Protocol - Analysis and Applications, J. A. Garay et al., EUROCRYPT2015LNCSVOI9057, Vol. II, pp. 281310 (eprint.iacr.org/2016/1048)

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1AM302) EXPLAINABLE AI

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce the concepts of explainable AI and interpretable machine learning
- To understand concepts of black-box machine learning, explainable AI methods
- To analyse various explainable AI methods
- To understand various applications of explainable deep learning
- To identify various challenges related to explainable AI

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the context of the machine learning applications with respect to explain ability and interpretability

CO-2: Analyze concepts of various ML model validation, evaluation, and performance visualization for both supervised and unsupervised learning

CO-3: Apply state of the art explainable AI tools, techniques and interpretable methods for various AI applications

CO-4: Correlate explainable deep learning techniques and its applications

CO-5: Interpret various XAI applications and future challenges related to Explainable AI

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	1	1	1	1	-	-	1	1	2	2	3	1	1	1
CO-2	2	2	2	1	1	-	1	1	1	2	2	3	2	2	3
CO-3	3	2	3	2	2	1	1	2	2	3	3	3	2	2	3
CO-4	3	1	2	3	2	2	1	1	2	3	3	3	3	2	2
CO-5	3	2	1	2	3	2	-	1	2	3	3	3	3	3	2

UNIT-I:

Introduction: Black-Box problem, Goals, Brief History, Purpose, Societal Impact, Types of Explanations, Trade-offs, Taxonomy, Flowchart for Interpretable and Explainable Techniques.

Pre-model Interpretability and Explainability: Data Science Process and EDA, Exploratory Data Analysis, Feature Engineering.

UNIT-II:

Model Visualization Techniques and Traditional Interpretable Algorithms: Model Validation, Evaluation, and Hyperparameters, Model Selection and Visualization, Classification Model Visualization, Regression Model Visualization, Clustering Model Visualization, Interpretable Machine Learning Properties, Traditional Interpretable Algorithms

UNIT-III:

Model Interpretability: Advances in Interpretable Machine Learning: Interpretable vs. Explainable. Algorithms, Tools and Libraries, Ensemble-Based, Decision Tree-Based, Rule-Based Techniques, Scoring System.

Post-Hoc Interpretability and Explanations: Tools and Libraries, Visual Explanation, Feature Importance, Example-Based.

UNIT-IV:

Explainable Deep Learning: Applications, Tools and Libraries, Intrinsic, Perturbation, Gradient / Backpropagation.

UNIT-V:

Explainability: Time Series Forecasting, Natural Language Processing, and Computer Vision XAI: Challenges: Properties of Explanation, Categories of Explanation, Taxonomy of Explanation Evaluation.

XAI: Future: Formalization of Explanation Techniques and Evaluations, Adoption of Interpretable Techniques, Human-Machine Collaboration, Collective Intelligence from Multiple Disciplines, Responsible AI (RAI), XAI and Security, Causality and XAI.

TEXT BOOK:

1. Explainable Artificial Intelligence: An Introduction to Interpretable Machine Learning, Uday Kamath and John Liu, 1st Edition, Springer, 2021

REFERENCES:

1. Explainable AI with Python, Leonida Gianfagna and Antonio Di Cecco, 1st Edition, Springer International, 2021
2. Hands-On Explainable AI (XAI) with Python, Denis Rothman, 1st Edition, Packt Publishing, 2020

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1DS405) TIME SERIES ANALYSIS AND FORECASTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To learn basic analysis of time series data and time series regression
- To learn auto-regressive and model averaging models
- To learn basic concepts of forecasts for a time series data
- To learn how to perform the advanced computation like Multivariate Time Series models and forecasting

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Apply ideas to real time series data and interpret outcomes of analyses

CO-2: Learn how to construct a time-series plot and identifying the underlying patterns in the data

CO-3: Learn basic concepts factorization techniques in time series analysis

CO-4: Learn how to develop forecasts for a time series that has a seasonal pattern

CO-5: Demonstrate an advanced understanding of the underlying concepts in the time series and frequency domains for computation, visualization, and analysis of time series data

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	3	3	2	-	-	2	2	2	3	3	2	1
CO-2	3	2	3	2	3	3	-	-	2	2	2	3	2	1	2
CO-3	3	2	3	3	3	3	-	-	3	2	3	2	3	2	2
CO-4	3	2	2	3	2	2	-	-	3	2	2	3	3	2	3
CO-5	3	2	2	3	2	3	-	-	2	2	3	2	3	2	3

UNIT-I:

Introduction of time series analysis, The Nature of Time Series Data, Internal structures of time series, Time Series Statistical Models, Moving Window for Time series data.

UNIT-II:

Fourier decomposition, deep learning features LSTM, Image Histogram, Deep learning feature CNN, Relational data, Graph data, Indicator Variable, Feature binning, interaction variable.

UNIT-III:

Mathematical transforms, Model specific featurizations, Feature orthogonality, Domain specific featurizations, Feature slicing.

Kaggle Winner Solutions: Students are suggested to carry out with at least 3 Kaggle winnercompetition case study solutions and present the report for that.

(<https://www.kaggle.com/code/sudalairajkumar/winning-solutions-of-kaggle-competitions>)

UNIT-IV:

Autocorrelation and Partial autocorrelation, General Approach to Time Series Modelling and Forecasting, Evaluating and Monitoring Forecasting Model Performance, Introduction Least Squares Estimation in Linear Regression Models, Statistical Inference in Linear Regression.

UNIT-V:

Model Adequacy Checking, Regression Models for General Time Series Data. Exponential Smoothing, First order and Second order, Multivariate Time Series Models and Forecasting, Multivariate Stationary Process, ARIMA model, Vector AR (VAR) Models.

TEXT BOOKS:

1. Introduction to Time Series Analysis and Forecasting, Douglas C. Montgomery, Cheryl L. Jen, 2nd Edition, Wiley Series in Probability and Statistics, 2015
2. Master Time Series Data Processing, Visualization, and Modeling Using Python, Dr. Avishek Pal, Dr. P. K. S. Prakash, 2017

REFERENCES:

1. Introduction to Time Series Analysis, P. J. Brockwell and R. A. Davis
2. Time Series Analysis and Its Applications, Robert H. Shumway and David S. Stoffer
3. Introduction to Statistical Time Series, W. A. Fuller
4. Time Series Analysis, Wilfredo Palma

B.Tech. VIII Semester

(22PE1DS406) QUANTUM COMPUTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the basics of quantum computation and quantum mechanics
- To apply different models of quantum computation
- To illustrate the quantum models using machine learning algorithms
- To explore various Quantum algorithms

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Explore the quantum computation environment

CO-2: Identify the implications of quantum computation in the field of mathematics and associated fields

CO-3: Demonstrate building blocks and programming methodologies for quantum computing

CO-4: Apply quantum computation algorithms

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	3	2	-	-	2	2	2	3	3	2	2
CO-2	2	2	3	2	3	3	-	-	2	2	2	3	2	2	2
CO-3	2	2	3	3	3	3	-	-	3	2	3	2	2	2	2
CO-4	3	2	2	3	2	2	-	-	3	2	2	3	3	2	3

UNIT-I:

Introduction and Background: Overview, The Circuit Model of Computation, A Linear Algebra Formulation of the Circuit Model, Reversible Computation. A Preview of Quantum Physics.

Linear Algebra and The Dirac Notation: The Dirac Notation and Hilbert, Dual Vectors, Operators, The Spectral Theorem, Functions of Operators

UNIT-II:

Qubits and The Framework of Quantum Mechanics: The State of a Quantum System, Time-Evolution of a Closed System, Composite Systems.

A Quantum Model of Computation: The Quantum Circuit Model, Quantum Gates, 1-Qubit Gates, Controlled-U Gates, Universal Sets of Quantum Gates, Efficiency of Approximating Unitary Transformations.

UNIT-III:

Superdense Coding and Quantum Teleportation: Superdense Coding, Quantum Teleportation, An Application of Quantum Teleportation.

Quantum Error Correction: Error Models for Quantum Computing, Encoding, Error Recovery. Architecture: Bits and Qubits, Classical Gates, Reversible Gates, Quantum Gates.

UNIT-IV:

Programming Languages: Programming in a Quantum World, Quantum Assembly Programming, Toward Higher-Level Quantum Programming.

Theoretical Computer Science: Deterministic and Nondeterministic Computations, Probabilistic Computations, Quantum Computations.

UNIT-V:

Quantum Algorithms: Deutsch's Algorithm, The Deutsch–Jozsa Algorithm, Simon's Periodicity Algorithm, Grover's Search Algorithm, Shor's Factoring Algorithm

TEXT BOOKS:

1. An Introduction to Quantum Computing, Phillip Kaye, Raymond Laflamme, Michele Mosca, Oxford University Press
2. Quantum Computing for Computer Scientists, Noson S. Yanofsky, Michael A. Mannucci, Cambridge University Press

REFERENCES:

1. Quantum Computing, Ronald De Wolf, QuSoft, CWI and University of Amsterdam
2. Quantum Computing Explained, David McMahon, Wiley
3. Quantum Computing Fundamentals, William (Chuck) Easttom

B.Tech. VIII Semester

(22PE1AM407) ROBOTIC PROCESS AUTOMATION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To explore the knowledge of basic concepts of Robotic Process Automation
- To outline the RPA design & development strategies
- To apply design methodologies adopted by UiPath
- To develop flow diagrams using RPA for real-world application
- To apply programming robots for logging and exception handling

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand Robotic Process Automation concepts

CO-2: Adopt UiPath programming techniques to deploy robot configurations

CO-3: Explore various data extraction techniques and perform integrations with various popular applications

CO-4: Design and develop a programmed robot that includes logging and exception handling

CO-5: Deploy and control Bots with UiPath orchestrator

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	1	1	3	3	2	2	2	1	1	2	2	2	2	1
CO-2	2	2	2	1	3	1	1	2	1	1	1	1	3	2	2
CO-3	1	2	2	3	2	2	1	3	1	2	1	2	3	2	3
CO-4	2	2	3	3	2	2	1	2	1	2	1	2	2	2	3
CO-5	1	2	1	1	3	1	1	2	1	2	1	1	2	3	3

UNIT-I:

Introduction: What is Robotic Process Automation (RPA), Scope & techniques of Automation, Benefits of RPA, Components of RPA, RPA Platforms, UiPath Studio, Installation of UiPath Studio, Learning UiPath Studio.

UNIT-II:

Sequence, Flowchart & Control Flow: Sequencing the Workflow, Activities, Flowchart, Control Flow for Decision making. Data Manipulation: Variables, Collection, Arguments, Data Table, Clipboard management, File operations with step-by-step example

Controls: Finding the control, waiting for a control, Act on a control, UiExplorer, Handling Events Recording and advanced UI Interaction Basic recording, Desktop recording, Web recording, Citrix, Screen Scraping, when to use OCR, Types of OCR available, how to use OCR Avoiding typical failure points.

UNIT-III:

Plugins and Extensions: Terminal plugin, Java plugin, Java plugin with UiPath Studio, Citrix automation, Citrix environment, Mail plugin, PDF plugin, Web integration, Excel and Word plugins, Credential management Extensions Handling User Events and Assistant Bots: What are assistant bots, Monitoring system event triggers: Hotkey trigger, Mouse trigger, System trigger, Monitoring image and element triggers, Launching an assistant bot on a keyboard event.

UNIT-IV:

Exception Handling, Debugging, and Logging: Exception handling, Common exceptions and ways to handle them, Logging and taking screenshots Debugging techniques, setting break points, Slow step, Highlighting, Break, collecting crash dumps: Enabling crash dumps, Disabling crash dumps, Error reporting: Enterprise Edition customers, Community Edition users. Managing and Maintaining the Code: Project organization, picking an appropriate layout for each workflow, Breaking the process into smaller parts, Using exception handling, Making your workflow readable, Keeping it clean, Nesting workflows, Reusability of workflows, Templates, Commenting techniques, State Machine, When to use Flowcharts State Machines or Sequences.

UNIT-V:

Deploying and Maintaining the Bot: Publishing using publish utility, Overview of Orchestration Server, Using Orchestration Server to control bots, Using Orchestration Server to deploy bots, License management, Activating and uploading a license to Orchestrator, Publishing and managing updates, Packages, Managing packages.

TEXT BOOK:

1. Learning Robotic Process Automation, Alok Mani Tripathi, O'Reilly Publishing, 2018

REFERENCES:

1. Robotic Process Automation: Guide To Building Software Robots, Automate Repetitive Tasks & Become An RPA Consultant, Richard Murdoch, Amazon Asia-Pacific Holdings, 2018
2. Introduction to Robotic Process Automation: A Primer, Institute of Robotic Process Automation, Frank Casale, Rebecca Dilla, Heidi Jaynes, Lauren Livingston, Amazon Asia-Pacific Holdings, 2018

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CS405) SOCIAL MEDIA ANALYTICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Graph Theory, Data Mining, Python/R programming

COURSE OBJECTIVES:

- To familiarize with the concept of social media
- To familiarize with the concept of social media analytics and understand its significance
- To enable to develop skills required for analyzing the effectiveness of social media
- To examine the ethical and legal implications of leveraging social media data
- To familiarize with different visualization techniques for Social media analytics

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the concept of social media analytics and its significance

CO-2: Analyze the effectiveness of social media

CO-3: Use different effective visualization techniques to represent social media analytics

CO-4: Use different social media analytics tools effectively and efficiently

CO-5: Acquire the fundamental perspectives and hands-on skills needed to work with social media data

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	3	3	2	-	-	2	2	2	3	2	1	3
CO-2	3	2	3	2	3	3	-	-	2	2	2	3	3	1	2
CO-3	3	2	3	3	3	3	-	-	3	2	3	2	3	1	2
CO-4	3	2	2	3	2	2	-	-	3	2	2	3	3	1	2
CO-5	3	2	2	3	2	3	-	-	2	2	3	2	2	1	2

UNIT-I:

Social Media Analytics: An Overview: Core Characteristics of Social Media, Types of Social Media, Social media landscape, Need for Social Media Analytics (SMA), SMA in small & large organizations. Purpose of Social Media Analytics, Social Media vs. Traditional Business Analytics, Seven Layers of Social Media Analytics, Types of Social Media Analytics, Social Media Analytics Cycle, Challenges to Social Media Analytics, Social Media Analytics Tools

UNIT-II:

Social Network Structure, Measures & Visualization: Basics of Social Network Structure - Nodes, Edges & Tie Describing the Networks Measures - Degree Distribution, Density, Connectivity, Centralization, Tie Strength & Trust Network Visualization - Graph Layout,

Visualizing Network features, Scale Issues. Social Media Network Analytics - Common Network Terms, Common Social Media Network Types, Types of Networks, Common Network Terminologies, Network Analytics Tools.

UNIT-III:

Social Media Text, Action & Hyperlink Analytics: Social Media Text Analytics - Types of Social Media Text, Purpose of Text Analytics, Steps in Text Analytics, Social Media Text 8 Analysis Tools Social Media Action Analytics - What Is Actions Analytics? Common Social Media Actions, Actions Analytics Tools Social Media Hyperlink Analytics - Types of Hyperlinks, Types of Hyperlink Analytics, Hyperlink Analytics Tools

UNIT-IV:

Social Media Location & Search Engine Analytics: Location Analytics - Sources of Location Data, Categories of Location Analytics, Location Analytics and Privacy Concerns, Location Analytics Tools Search Engine Analytics - Types of Search Engines, Search Engine Analytics, Search Engine Analytics Tools

UNIT-V:

Social Information Filtering: Social Information Filtering - Social Sharing and filtering, Automated Recommendation systems, Traditional Vs social Recommendation Systems Understanding Social Media and Business Alignment, Social Media KPI, Formulating a Social Media Strategy, Managing Social Media Risks

TEXT BOOKS:

1. Seven Layers of Social Media Analytics - Mining Business Insights from Social Media Text, Actions, Networks, Hyperlinks, Apps, Search Engine, and Location Data, Gohar F. Khan, 2015
2. Mining the Social Web - Analyzing Data from Facebook, Twitter, LinkedIn, and Other Social Media Sites, Matthew A. Russell, O'Reilly
3. Social Network Data Analytics, Charu Aggarwal (Ed.), Springer, 2011

REFERENCES:

1. Analyzing the Social Web, Jennifer Golbeck, 1st Edition
2. Social Media Analytics, Techniques and Insights for Extracting Business Value Out of Social Media, Matthew Ganis, Avinash Kohirkar, IBM Press, 2015
3. Social Media Analytics Strategy - Using Data to Optimize Business Performance, Alex Gonçalves, APress
4. Social Media Data Mining and Analytics, Szabo G., G. Polatkan, O. Boykin & A. Chalkiopoulos, Wiley, 2019
5. Social Media Analytics: Effective Tools for Building, Interpreting and Using Metrics, Marshall Sponder, 1st Edition, McGraw-Hill, 2011

ONLINE RESOURCES:

1. <https://cse.iitkgp.ac.in/~pawang/courses/SC16.html>
2. https://onlinecourses.nptel.ac.in/noc20_cs78/preview